

Technical Lab Report: Splunk SIEM Monitoring & Threat Detection Lab

1. Executive Summary & Topology

This technical report details the end-to-end engineering process to deploy a fully functional **SIEM (Security Information and Event Management)** home lab environment using **Splunk Enterprise**.

The goal of this lab is to aggregate, parse, and alert on real-time Windows security events and system logs, establishing defensible detection engineering controls against offensive tactics simulated via an adversary platform.

Windows Server 2022 credentials:

Hostname: Jazon

Username: Administrator

Password: *****

Splunk Credentials:

Username: Jazon

Password: *****

Splunk Login Website: localhost:8000/en-US/account/login or just enter localhost:8000 to enter

Environment Architecture

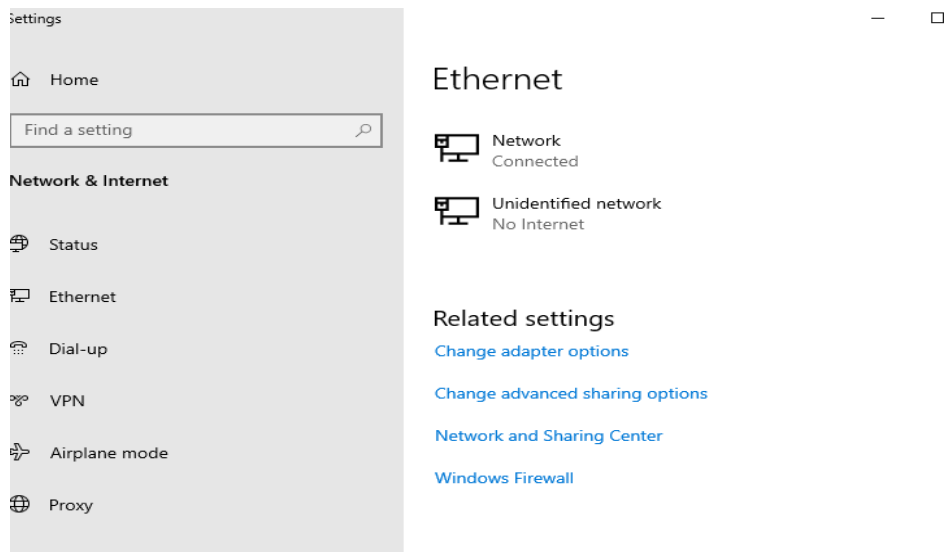
- **Hypervisor:** Oracle VM VirtualBox
- **SIEM Central Manager:** Windows Server 2022 (Standard / Evaluation)
 - **IP Address:** 192.168.1.50 (Static)
 - **Role:** Splunk Enterprise Indexer, Search Head, and Deployment Manager
- **Target Machine (Log Source):** Monitored locally on the Windows Server instance via Splunk Universal Forwarder + Sysmon.
- **Adversary Simulation Endpoint:** Kali Linux
 - **IP Address:** 192.168.1.60 (Static)
 - **Role:** Network scanning, brute-forcing, and vulnerability exploitation.

2. Network Topology & Static IP Engineering

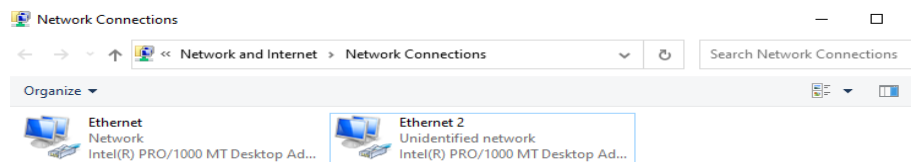
To prevent broken communication streams between data, ingest engines, persistent static assignments are required within a VirtualBox Host-Only Adapter or Bridged network segment.

A. Windows Server (192.168.1.50) Configuration

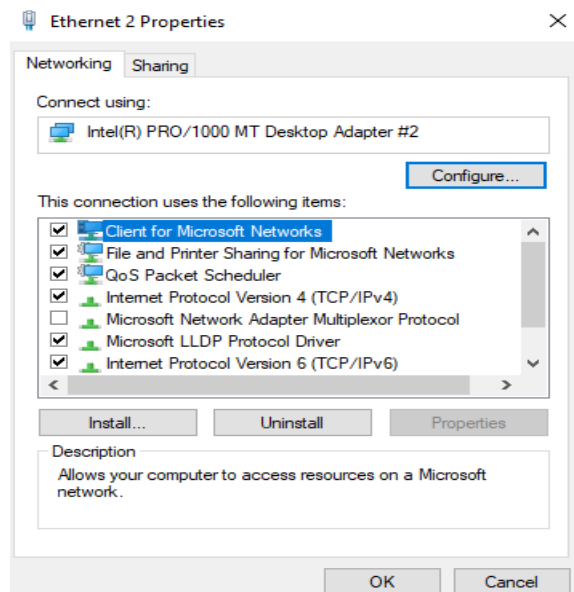
1. Log into Windows Server as an Administrator.
2. Type in Ethernet Settings in search bar
3. Select Adapter options



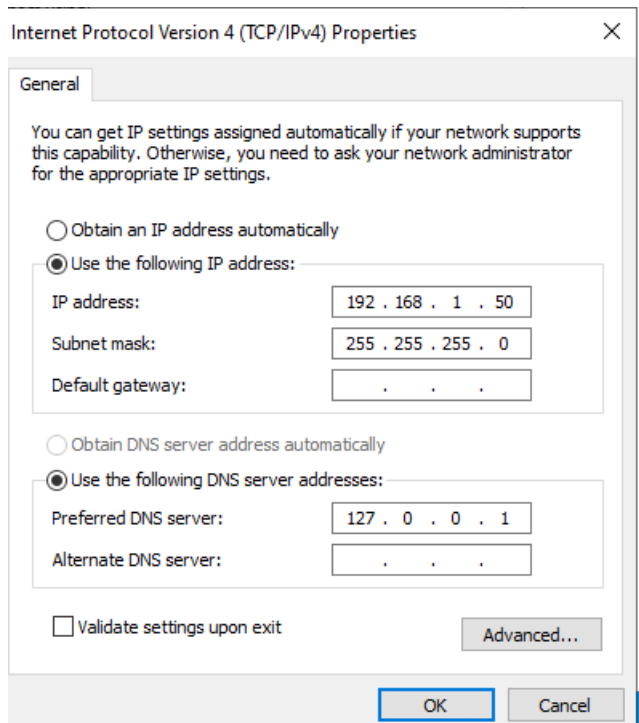
4. Right Click Ethernet 2 and select on properties



5. Double Click Internet Protocol Version 4



6. Enter the ip addresses into it



7. Once done select okay and exit the page
8. Validate settings via ipconfig /all in cmd

```
Ethernet adapter Ethernet:  
  
    Connection-specific DNS Suffix  . : phub.net.cable.rogers.com  
    Link-local IPv6 Address . . . . . : fe80::59b7:d797:bdbc:ae3%12  
    IPv4 Address. . . . . : 10.0.2.15  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . : 10.0.2.2  
  
Ethernet adapter Ethernet 2:  
  
    Connection-specific DNS Suffix  . :  
    Link-local IPv6 Address . . . . . : fe80::ec15:8a22:beea:4768%19  
    IPv4 Address. . . . . : 192.168.1.50  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . :  
  
C:\Users\Administrator>
```

B. Kali Linux (192.168.1.60) Configuration

1. Open a terminal terminal in Kali Linux..
2. Run the following command to assign the IP address and subnet mask (using /24 for 255.255.255.0):

sudo ip addr add 192.168.1.60/24 dev eth1
3. Bring the interface up (turn it on) by running

sudo ip link set eth1 up

4. Verify the change by checking the interface details:

ip addr show eth1

```
(kali@kali)-[~]
$ ip addr show eth1
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f9:2c:8d brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.60/24 brd 192.168.1.255 scope global noprefixroute eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::5389:5574:6ae6:c40a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

5. Enter ifconfig to look up all the ip addresses in the network

```
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::df9e:3e27:28f0:8279 prefixlen 64 scopeid 0<link>
    ether 08:00:27:ad:25:87 txqueuelen 1000 (Ethernet)
    RX packets 1 bytes 590 (590.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 27 bytes 3338 (3.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.60 netmask 255.255.255.0 broadcast 0.0.0.0
    ether 08:00:27:f9:2c:8d txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 56 bytes 5200 (5.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

3. Splunk Enterprise & Data Ingestion Setup

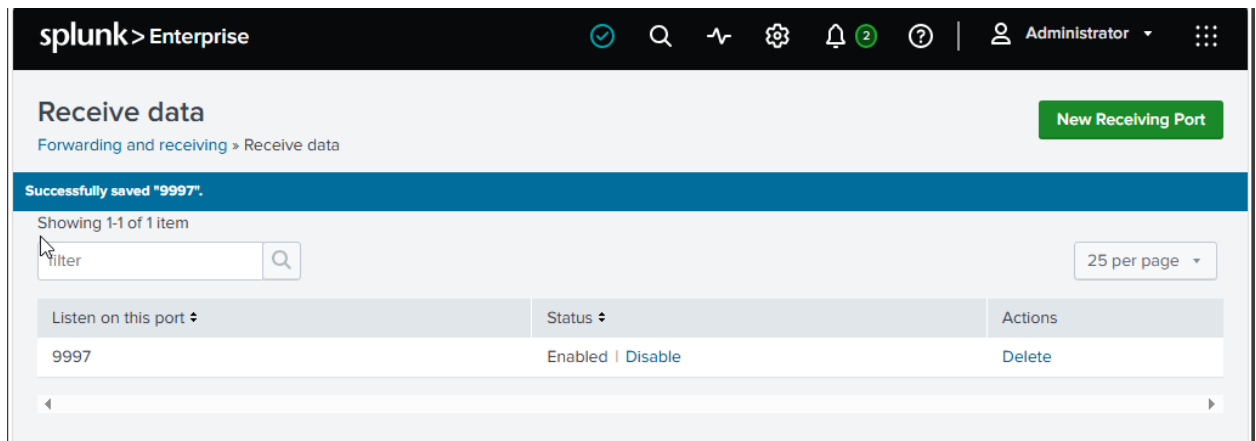
Step 1: Installing Splunk Enterprise on Windows Server

1. Download the Windows MSI installer package (splunk-X.X.X-x64-release.msi) from the official Splunk platform.
2. Run the installer
3. Open a browser and access the control panel at <http://localhost:8000> (or <http://192.168.1.50:8000>). Log in with the configured administrative credentials.

Step 2: Configure Network Network Receivers

Splunk Enterprise must have its TCP network data parsing daemon actively listening to receive logs from agents.

1. In the top right ribbon of Splunk Web, go to **Settings > Forwarding and receiving**.
2. Click **Configure receiving > New Receiving Port**.
3. Specify port 9997 and save changes.



Step 3: Implement Advanced Endpoint Auditing (Sysmon)

Standard Windows Event Logs miss critical process execution telemetry. To fix this, deploy Microsoft Sysinternals Sysmon:

1. Download [Microsoft Sysmon](#) and the [SwiftOnSecurity Sysmon Config Template](#).
2. Execute the setup with the configuration mapping file:

DOS

```
sysmon64.exe -i sysmonconfig.xml
```

Step 4: Installing the Splunk Universal Forwarder

1. Download the Splunk Universal Forwarder installer to the server.
2. Execute a silent configuration routine to index native security logs alongside Sysmon data streams:
3. Point the Forwarder explicitly to look at Sysmon by creating

```
Receive Data | "Untitled - Notepad"
Select Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> cd 'C:\Program Files\SplunkUniversalForwarder\etc\system\local\'
PS C:\Program Files\SplunkUniversalForwarder\etc\system\local> ls

Directory: C:\Program Files\SplunkUniversalForwarder\etc\system\local

Mode                LastWriteTime         Length Name
----                -
-a----          9/15/2026  11:47 AM             126 authentication.conf
-a----          9/15/2026  11:48 AM             152 outputs.conf
-a----          8/19/2026   5:33 PM             273 README
-a----          9/15/2026  11:49 AM             458 server.conf
-a----          9/15/2026  11:48 AM              48 web.conf
```

```
PS C:\Program Files\SplunkUniversalForwarder\etc\system\local> @'
>> [WinEventLog://Microsoft-Windows-Sysmon/Operational]
>> disabled = false
>> renderXml = true
>> index = wineventlog
>> sourcetype = XmlWinEventLog:Sysmon
>> '@ | Out-File -FilePath "C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf" -Encoding utf8 -Append
PS C:\Program Files\SplunkUniversalForwarder\etc\system\local> ls

Directory: C:\Program Files\SplunkUniversalForwarder\etc\system\local

Mode                LastWriteTime         Length Name
----                -
-a----          9/15/2026  11:47 AM             126 authentication.conf
-a----          9/15/2026  12:02 PM             146 inputs.conf
-a----          9/15/2026  11:48 AM             152 outputs.conf
-a----          8/19/2026   5:33 PM             273 README
-a----          9/15/2026  11:49 AM             458 server.conf
-a----          9/15/2026  11:48 AM              48 web.conf

PS C:\Program Files\SplunkUniversalForwarder\etc\system\local> type "C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf"
[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = false
renderXml = true
index = wineventlog
sourcetype = XmlWinEventLog:Sysmon
PS C:\Program Files\SplunkUniversalForwarder\etc\system\local> net
```

- Restart the service: net stop SplunkForwarder && net start SplunkForwarder.

```
PS C:\Program Files\SplunkUniversalForwarder\etc\system\local> net stop SplunkForwarder

The SplunkForwarder service was stopped successfully.

PS C:\Program Files\SplunkUniversalForwarder\etc\system\local>
PS C:\Program Files\SplunkUniversalForwarder\etc\system\local> net start SplunkForwarder
The SplunkForwarder service is starting...
The SplunkForwarder service was started successfully.

PS C:\Program Files\SplunkUniversalForwarder\etc\system\local>
```

- Configure inputs to route Windows Event Logs and Sysmon operational logs. Edit or append to: C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf

PlainText:

[WinEventLog://Security]

disabled = 0

index = main

[WinEventLog://Microsoft-Windows-Sysmon/Operational]

disabled = false

renderXml = true

index = main

sourcetype = XmlWinEventLog:Sysmon

6. Restart the service: net stop SplunkForwarder && net start SplunkForwarder.

```
PS C:\Program Files\SplunkUniversalForwarder\etc\system\local> net stop SplunkForwarder
The SplunkForwarder service was stopped successfully.

PS C:\Program Files\SplunkUniversalForwarder\etc\system\local>
PS C:\Program Files\SplunkUniversalForwarder\etc\system\local> net start SplunkForwarder
The SplunkForwarder service is starting...
The SplunkForwarder service was started successfully.

PS C:\Program Files\SplunkUniversalForwarder\etc\system\local>
```

7. Verify active Connection Status:

./splunk list forward-server

4. Threat Detection & Log Analysis Scenarios

Test Scenario A: Active Reconnaissance (Nmap Target Sweep)

- Execution (From Kali):

```
(kali㉿kali)-[~]
└─$ sudo nmap -sS -p 445,3389 -Pn 192.168.1.50
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-15 15:15 EDT
Nmap scan report for 192.168.1.50
Host is up (0.0012s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds
3389/tcp   filtered  ms-wbt-server
MAC Address: 08:00:27:C8:72:F6 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.40 seconds
```

- **Splunk Detection Query (Search & Reporting):**

The screenshot shows the Splunk Enterprise web interface. The search bar contains the query: `index=* EventCode=5156 Source_Address=192.168.1.60 | stats count BY Destination_Port`. The results are displayed in a table with the following data:

Destination_Port	count
3702	1
445	4
8	4

Analysis Focus: Identifies immediate spikes in firewall permit validations across restricted infrastructure points.

Test Scenario B: Brute-Force Authentication Attack (RDP/SMB)

- **Execution (From Kali):**

Bash

```
hydra -l Administrator -P /usr/share/wordlists/rockyou.txt rdp://192.168.1.50
```

```

kali@kali:~/usr/share/wordlists$
hydra -l Administrator -P /usr/share/wordlists/rockyou.txt rdp://192.168.1.50
Hydra v2.5 (C) 2023 by van Hauser/THC - David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-26 11:02:06
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3 to wait between connection to allow the server to recover
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 4 tasks per 1 server, overall 4 tasks, 1034599 login tries (111/p:1434399), ~358100 tries per task
[DATA] attacking rdp://192.168.1.50:3389/
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: password, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 12345, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 123456789, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 123456, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 1234567, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: rockyou, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: princess, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 1loveyou, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 12345678, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: abc123, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: nic[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Admini
strator password: daniel, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: babygirl, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: jessica, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: lovely, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: monkey, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 654321, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: michael, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: qwe[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Admini
strator password: ashley, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 111111, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 11[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Admini
strator password: michelle, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: 000000, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: sunshine, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: password, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: tiger, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: chocolate, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: anthony, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: but[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Admini
strator password: soccer, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: friends, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: purple, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: jordan, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: liverpool, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: angel, continuing attacking the account.
[3389][rdp] account on 192.168.1.50 might be valid but account not active for remote desktop: login: Administrator password: fuckyou, continuing attacking the account.

```

• Splunk Detection Query:

Plaintext

Search | Splunk 10.4.0

Not secure 192.168.1.50:8000/en-US/app/search/search?earliest=0&latest=&q=search%20index%3D%20Even...

splunk > Enterprise

New Search

index=* EventCode=4625

Time range: All time

✓ 4 events (before 9/24/26 8:10:06.000 AM) No Event Sampling

Events (4) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 day per column

Format Show: 20 Per Page View: List

< Hide Fields All Fields

Dashboards

source 1

sourcetype 1

INTERESTING FIELDS

Account_Domain 2

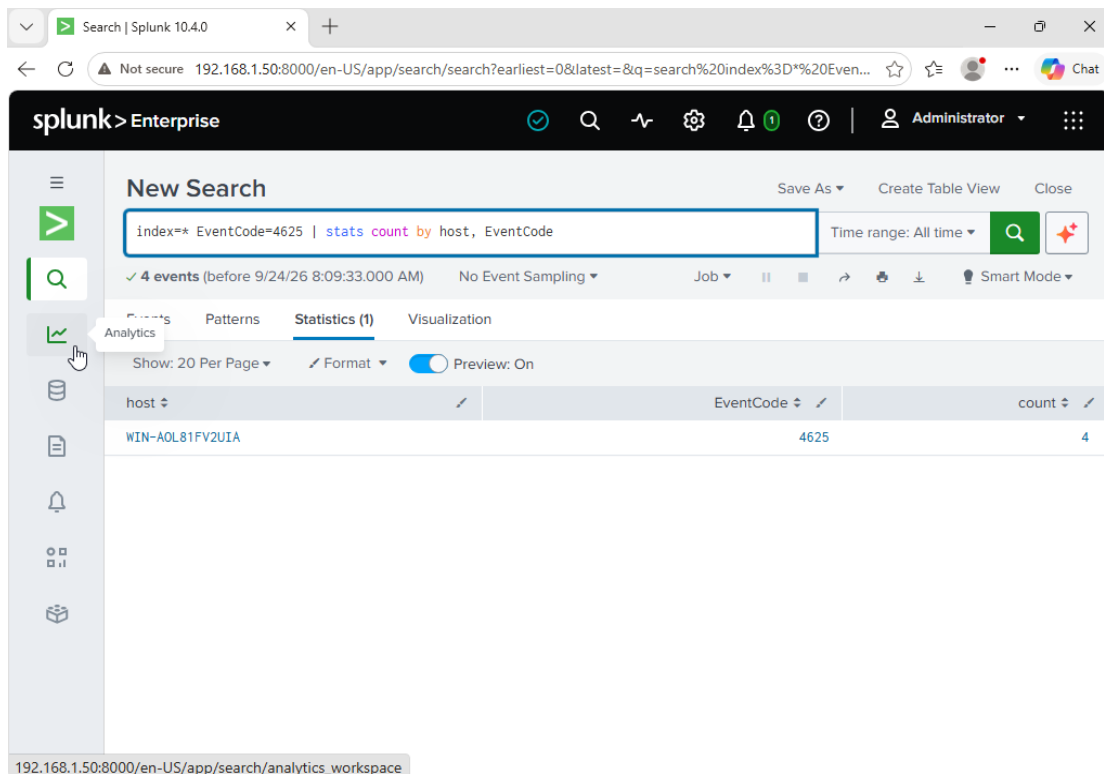
Account_Name 3

Authentication_Package 1

Caller_Process_ID 4

Time	Event
9/23/26 10:15:46.157 AM	LogName=Security EventCode=4625 EventType=0 ComputerName=WIN-AOL81FV2UIA Show all 61 lines
8/31/26 08:38:13.220 AM	LogName=Security EventCode=4625

192.168.1.50:8000/en-US/app/search/dashboards



Analysis Focus: Isolates massive volumes of event ID 4625 (An account failed to log on) within short anomalies.

5. Comprehensive Troubleshooting Framework

Phase 1: Local OS Telemetry Failures

1. Inbound Packets Blocked / Suppressed

- **The Error:** Active Windows Defender Firewall or Network Inspection System silently drops reconnaissance scan sequences (nmap), rendering ports completely invisible or falsely reported as filtered/closed.
- **The Fix:** Disable local firewall profiles and real-time monitoring temporarily via PowerShell to isolate raw operating system log generation:

PowerShell:

Set-NetFirewallProfile -Profile Domain,Private,Public -Enabled False Set-MpPreference -DisableRealtimeMonitoring \$true

2. Missing Windows Filtering Platform (WFP) Logs

- **The Error:** Standard Windows Security Logs do not natively generate telemetry for allowed inbound network connection attempts (Event ID 5156). Running an nmap sweep yields zero security event results.

- **The Fix:** Enable advanced security auditing policies locally for Filtering Platform connections:

```
auditpol /set /subcategory:"Filtering Platform Connection" /success:enable /failure:enable
gpupdate /force
```

- **Verification Metric:** Confirm log ingestion locally via PowerShell:
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=5156; StartTime=(Get-Date).AddMinutes(-10)} -ErrorAction SilentlyContinue

Phase 2: Splunk Data Transport & Network Failures

3. Missing Splunk Indexer Port Binding

- **The Error:** The Splunk Universal Forwarder remains inactive or throws errors when listing forward-servers.
- **The Cause:** Splunk Enterprise does not open its TCP indexing daemon by default.
- **The Fix:** Enable receiving port 9997 via Splunk Web UI (Settings > Forwarding and receiving > Receive data) and verify listening state in PowerShell:

PowerShell:

```
Get-NetTCPConnection -LocalPort 9997
```

4. Splunk Service Daemon Locks (splunkd)

- **The Error:** Attempting to restart SplunkForwarder produces a terminal exception: Cannot stop SplunkForwarder service on computer '.
- **The Cause:** Background splunkd process thread locks during active handshake retries (AutoLoadBalancedConnectionStrategy).
- **The Fix:** Terminate hung processes before cycling the service:

PowerShell:

```
Stop-Process -Name splunkd -Force
```

```
Start-Service -Name SplunkForwarder
```

5. Splunk Missing Index Target / Dropped Events

- **The Error:** Event ID 5156 appears in Windows Event Viewer, transport logs show Connected to idx=192.168.1.50:9997, but Splunk UI returns zero search results.
- **The Cause:** By default, the Universal Forwarder routes logs to an index named windows. If that index does not exist in Splunk Enterprise, logs are silently dropped.

- **The Fix: Explicitly direct telemetry to the standard main index inside C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf:**

[WinEventLog://Security]

disabled = 0

index = main

6. Curated Reference Materials & Infrastructure Repositories

Infrastructure Binaries

- [Splunk Enterprise Download Portal](#) – Primary analytics platform tracking asset.
- [Microsoft Sysinternals Suite](#) – High-fidelity OS activity collection.

Engineering & Implementation Manuals

- [Splunk Deployment & Security Architecture Setup](#) – Detailed configuration framework strategies.
- [SwiftOnSecurity Endpoint Monitoring Standards](#) – Production-grade event definitions mapping file.
- [Blue Team Labs / Splunk Fundamentals Basics](#) – Step-by-step visual references for deploying Windows Universal Forwarders securely.